

DBMS Semester Examination 2024

Database Management Systems (DBMS) · Semester Examination · 2024

Max. Marks	75	Time	3 hours
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Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Discuss the role of Database System Architecture in Introduction & ER Model. (5 marks)
2. Discuss the role of Relational Algebra in Relational Model & SQL. (5 marks)
3. Illustrate Functional Dependencies with a worked example. (5 marks)
4. Define ACID Properties. Derive or justify the key properties involved. (5 marks)
5. Discuss the role of File Organization in Storage, Indexing & Recovery. (5 marks)

Part B

1. Define Database System Architecture. Derive or justify the key properties involved. (Unit 1)
2. Define Data Independence. Derive or justify the key properties involved. (Unit 1)
3. Write short notes on Relational Algebra. (Unit 2)
4. Define Joins. Derive or justify the key properties involved. (Unit 2)
5. Illustrate Functional Dependencies with a worked example. (Unit 3)
6. Write short notes on Closure of Attributes. (Unit 3)
7. What is meant by ACID Properties? Discuss its advantages and limitations. (Unit 4)
8. Compare Serializability with Isolation Levels. When would you prefer one over the other? (Unit 4)
9. Define File Organization. Derive or justify the key properties involved. (Unit 5)
10. Compare B+ Tree Indexes with Log-Based Recovery. When would you prefer one over the other? (Unit 5)